

ments to follow the retrofit approach, where they retrofit a battery kit into an ICE vehicle and turn it into an EV. The biggest challenge right now is to build an EV from the ground up.

“You have to design the entire thing with a clean chit approach. This is where your knowledge of vehicle engineering, vehicle architecture, and the whole way of packaging it together in the most efficient and safe way is going to be tested, and this is where these technologies will be used,” notes Mukherjee.

Digitalisation ought to help in EV development. Yet, keeping abreast with these technologies can be cumbersome. Batteries, for instance, are constantly evolving with new chemistries and materials being tested every day. Additionally, keeping up with the hardware trends in terms of motors and powertrains is equally important.

“This is where you got to keep up, and this is where technologies like ML/AI will come in handy, wherein huge amounts of data points will come in handy and they can be put together so that you don’t waste time in doing hits and trials,” adds Mukherjee.

Lack of data points

There is more reliance in the EV segment, compared to the ICE, on digital product validation. The reason is that not all EV platforms are time-tested compared to the ICE engines.

If you look at EVs, by and large, the vehicle structure has not changed; instead of ICE powertrain, the EV powertrain has been embedded. So, whatever product development has been happening in the ICE related to structures will remain the same.

Speakers



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But for aspects like battery thermal management and other electronics being tested, there is no time-tested data available for EVs. Therefore simulation and data generation through IoT is even more essential to help innovate in the field.

Standardisation and academia-industry collaboration

“We should not miss the bus of this technological revolution like we did in case of ICE and then semiconductors. We should be more proactive and come up with a big portfolio of indigenous technology,” says Praveen Kumar, Professor at IIT-Guwahati.

But indigenous technology requires standardisation, and standardisation requires research. There is an urgent need for the academia to work on building innovations that are suitable for the Indian context and join hands with the industry to apply them as standards for the entire manufacturing and supply-

chain. “We need indigenous research instead of following what other countries are doing and then build EVs that fit the bill,” adds Mishra.

This will lay the groundwork for an application-wise dominion that is fixed right from the design stage, so any product falling into that application is always standardised irrespective of different manufacturers and the supply-chain. This will not only help in cost reduction but also decrease the lead time of the product reaching the market, eventually boosting EV adoption in the country.

The future of manufacturing

We are witnessing an exciting amalgamation of digital with hardware, making it one of the most exciting times for technology and manufacturing, which will have a profound impact over the next decade.

Mukherjee says, “We will see the entire energy mix change, and it is a game where multiple stakeholders will have to come together. I wish I was 25 years younger so I could participate more actively in this revolution!”

It is surely exciting to see how humanity will move away from fossil fuels to a more sustainable means of harnessing renewables and using it for both mobility and energy. With the right combination of working with academic institutions, OEMs, and consumers, which in today’s world is possible with the advent of internet connectivity, EV manufacturing will not only be revolutionised but will propel India to the global table of energy efficiency that it has targeted to meet by 2050. **EFY**

This article is based on opinions shared by the speakers during public events. Transcribed and curated by EFY’s Siddha Dhar.